

Homework (Habanero)

Open Ended Questions (FRQ)

Q9. Find the inverse of the function $f(x) =$

$\frac{1}{x+1}$. Verify that $f^{-1}(f(x)) = x$ and explain the result in the context of a medical dosage problem.

Q10. The growth rate of a certain type of cell is modeled by the function $G(t) = 100 \cdot (2)^t$, where t is the time in days. Find $G^{-1}(500)$ and interpret the result in the context of cell growth. Then, find $G(G^{-1}(500))$ and verify that it equals 500.

Chef's Tips

- Read each question carefully and choose the correct answer.
- Show all work for free response questions.
- Use a calculator only when necessary and be sure to show the calculation.